

Support vector machines

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A support vector machine (SVM) is a supervised machine learning (ML) method capable of learning from data and making decisions. The fundamental principles of the SVM were already introduced in the 1960s by Vapnik and Chervonenkis¹ in a theory that was further developed throughout the next decennia. However, it was only in the 1990s that SVMs attracted greater attention from the scientific community, and this was attributed to 2 significant improvements. The first extension is a *kernel trick* that allows the SVM to classify highly nonlinear problems.² The second permitted the extension of the SVM to solve problems in a regression framework³ called *support vector regression machine*. These improvements have resulted in a decisive general approximator that nowadays finds its use in many applications. Typically, the mathematics and theory behind SVMs are complex and require a deep understanding of optimization theory, algebra, and learning theory.

Nonetheless, the main idea can be intuitively explained, and this article will consider a classification problem to illustrate the concepts. In what follows, it can be noticed that SVMs differ from previously presented methods as they exploit geometries in the data and are not directly rooted in statistics (eg, generalized linear models). However, they originate from mathematics and engineering and are often compared with logistic regression explained in the previous article.

The starting point of an SVM is straightforward as it will try to solve a particular binary classification problem by the simplest model possible, separating the subjects that belong to the 2 different classes by a classification boundary. In 2 dimensions, this classification boundary will form a straight line. In 3 dimensions, this

classification boundary will become a plane, a line generalization. This boundary will be called a hyperplane for higher dimensions, which can be considered a plane in >3 dimensions and is beyond our imagination. Again, the question is how well such a simple model classifies and how well-learned concepts are generalizable to previously unseen data.

Figure 1, A shows a separable classification problem. It is perfectly possible to separate the *blue* from the *red* by using a straight line as a classification boundary. However, as illustrated in the plot, multiple options are possible. Which line should we select as our boundary to minimize the risk of misclassifying a previously unseen subject? The solution to this question presents itself in Figure 1, B and is called a *maximum-margin classifier*. The basic idea is simple. To minimize misclassification risk, we want our classification boundary positioned as far as possible from neighboring subjects belonging to the different classes. The margin is maximized between the classification boundary and the training data allowing for a tolerance region when predicting a class label for new subjects.

An important observation can be made from the figure. Data points far from the classification line do not influence its position. The only data points that determine the decision boundary are the 3 points in *black* in Figure 1, B. These points are called *support points* or *support vectors*. In other words, if we would remove all the subjects from our training dataset apart from these 3 support vectors, then the location of the decision boundary would remain unaltered. This example indicates that support vectors significantly influence the decision boundary, and changes in the training data will dramatically impact the decision boundary.

Figure 1, C shows an additional subject indicated by an *arrow* added to the training dataset. Coincidentally, this subject lies close to the decision boundary and is an influential support vector that will modify the maxim-margin problem, resulting in a different classification boundary, as indicated by the *green*. However, when eyeballing the classification problem, we can be pretty satisfied with the previous classifier, indicated by *dashes*, which yields wider overall margins concerning their neighboring training subjects.

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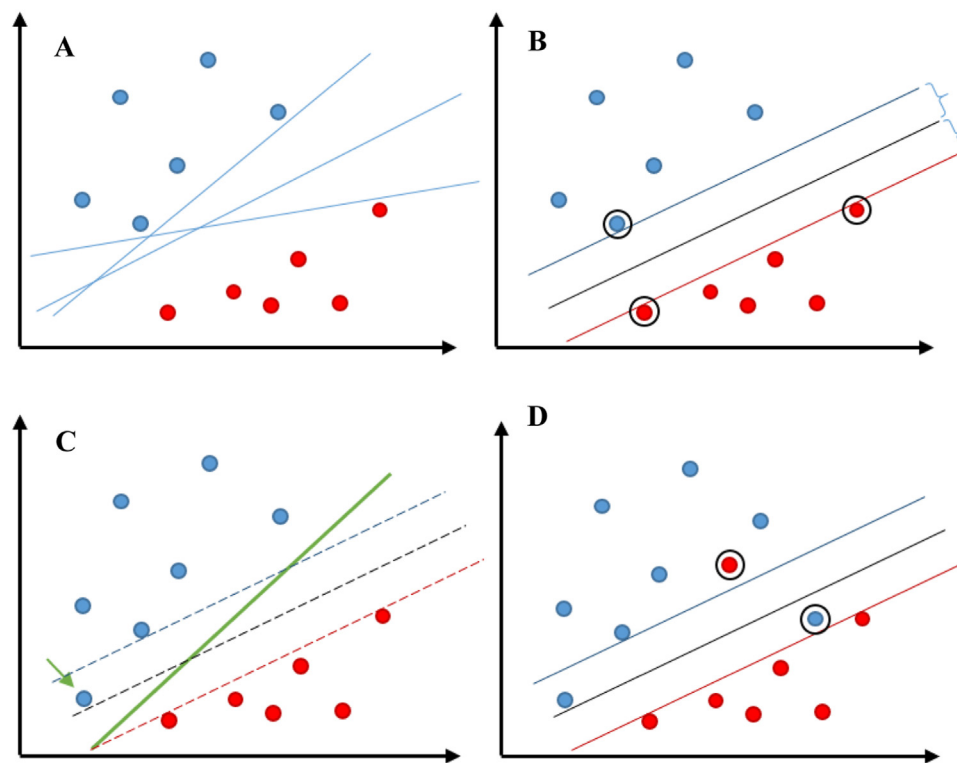


Fig 1. A and B illustrate the principle of the maximum-margin classifier; C and D show the introduction of the slack variable to allow the support vector classifier to maximize its margin while disregarding the influence of nearby observations even when the data is inseparable.

From the previous articles in this series on ML, we can recognize such behavior as a symptom of *overfitting* the training data. We want to avoid this overfitting by allowing some support vectors within the maximum margin, thus relaxing their influence on the decision boundary. A trade-off similar in spirit to the least absolute shrinkage and selection operator regression, discussed in a previous article,⁴ presents itself. In the case of SVMs, a budget-and-penalty system, called *slack variables*, is proposed to relax some of the support vectors. A fortiori, the same procedure can be used to pay for the penalty even when misclassifying subjects from the training dataset. For example, in the case of nonseparable data, as indicated in Figure 1, D, misclassifying the subjects encircled in black is unavoidable and tolerated by this budget-and-penalty system.

Up to this point, we only have considered simple classification problems, but how will an SVM perform on a complex nonlinear problem, such as the one displayed in Figure 2, A (showing that the decision boundary should be a circle)? However, with SVMs, we are restricted to straight lines, planes, or hyperplanes. The curse of dimensionality may offer a potential solution,

as previously discussed.⁴ The curse of dimensionality states that any classification problem can be perfectly solved when adding enough covariates such that the dimension of the dataset is much increased, allowing for a clear separation of subjects belonging to different classes. However, the example in Figure 2 has only 2 covariates or dimensions, so the SVM has to develop additional virtual dimensions. Toward this aim, the SVM will consider mapping the data onto a higher dimension, which is the opposite of the projection of high-dimensional data objects to a lower dimension.⁵ For example, consider a projection of the 2-dimensional data onto a spherical object as indicated in Figure 2, B. By this projection, the problem becomes separable with a simple plane that maximizes the margin with respect to neighboring subjects, as indicated by black. This mapping to higher dimensions is called a *kernel function*. The good news is that this mapping or projection does not need to be explicitly performed, as it is a computationally expensive step. Therefore, this step is often called the *kernel trick* because the SVM can find the maximum-margin classifier in an unspecified higher dimension. Note that the decision boundary in Figure 2,

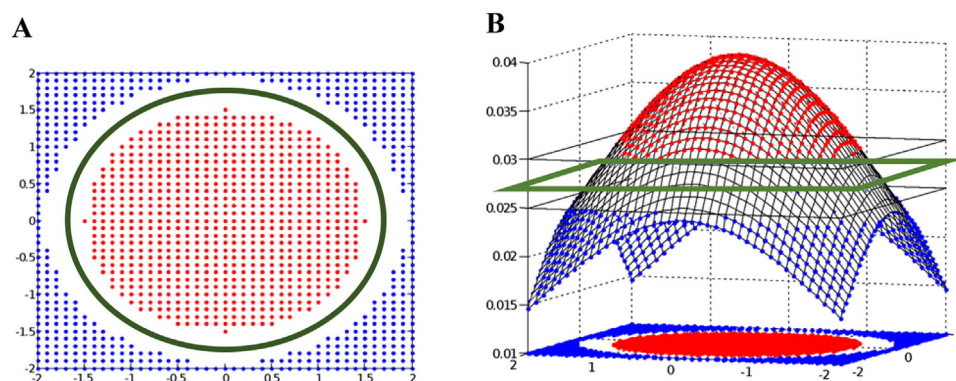


Fig 2. **A**, After back-transforming the hyperplane, the decision boundary is no longer linear in shape as indicated by the *green*; **B**, Illustration of a support vector machine that maps the binary classification data onto a higher dimension by a kernel trick such that separation by a hyperplane is achievable, as indicated by *green*.

B , is planar; when back-transformed to the original data space, the border will take a circular form or sometimes more complicated nonlinear boundaries.

Let us consider the classification problem introduced by Konstantonis et al⁶ to predict the extraction and non-extraction treatment on the basis of demographic, cephalometric, and model covariates. Other model-building parameters besides the parameters or coefficients that define the decision boundary must be decided on and estimated from the training data. These other model options are typically called *hyperparameters* and must be fine-tuned to select the most optimal and best-performing model. This practical example considers kernels (ie, linear, quadratic, cubic, and Gaussian) and optimizes over a grid of slack variables and kernel sizes using 5-fold cross-validation. This fine-tuning has resulted in a linear SVM with 1.3 on the slack variables budget. The resulting error table (often called *confusion matrix*) based on the 5-fold cross-validation is displayed in the [Table](#).

An error table is an informative summary of the performance of a classifier. The rows in this table indicate the ground truth values used as training labels, whereas the columns show how the SVM has classified the subject. A few things can be learned from such an error table. First of all, it can be noticed that 1 subject has been left out of the classification as the first row only sums up to 396 subjects. Missing values did motivate the removal of that subject from the study.

Furthermore, the classification accuracy can be computed by summing the diagonal cells (correct prediction) and dividing this number by the total sum of the cells ($[378 + 86]/[378 + 18 + 59 + 86]$), which gives an accuracy of 85.77%. The misclassification error

Table. Error table or confusion matrix of the SVM classifier from Konstantonis et al⁶

True class	Predicted class	
	Nonextraction	Extraction
Nonextraction	378	18
Extraction	59	86

can be computed similarly, but now by considering the off-diagonal cells (incorrect prediction), yielding an error of 14.23%. These classification results are reasonably good, but the error table provides us with more insight into the behavior of our SVM classifier. Note that for the largest class in this study (nonextraction), the SVM performed well; however, for extraction, the intraclass performance is weak as only half of the subjects in this class were correctly classified by our selected classifier. To summarize this intraclass performance, we use the concepts of sensitivity and specificity. Sensitivity is computed as the proportion of the correctly classified subjects that need an extraction divided by the total number of ground truth extractions in the dataset ($86/[59 + 86] = 59.31\%$), and the specificity is computed as the proportion of correctly classified subjects that do not require treatment divided by the total nontreated subjects in the dataset ($378/[378 + 18] = 95.45\%$). These numbers indicate that the classifier has difficulties correctly classifying the treatment cases. Further investigation is required to clarify the reason for this imbalance. Alternatively, a trade-off can be proposed to improve the intraclass performance at the cost of the overall classification accuracy.

To conclude, support vector machines are a powerful addition to the arsenal of ML methods, primarily because of the kernel trick. One could argue that projecting the data into unspecified higher dimensions is a bad practice in terms of overfitting the data, but the simple hyperplane model combined with a budget-penalty system and sufficient training observations should prevent large generalization errors. Not surprisingly, the linear SVM classifier was found as the optimal model on the Konstantonis dataset,⁶ avoiding overfitting as the many covariates (36) and limited observations (541) already yield a high-dimensional and sparse data space not requiring a mapping to higher dimensions. As mentioned earlier, SVMs are general approximators well-suited for prediction and classification tasks. However, when focusing on understanding the relationship between the outcome and the treatment predictors, SVMs are less suitable as no association model between the covariates and the outcome is constructed, as with generalized linear models.⁷ What is delivered by the SVM is only the notion of a support vector, which are subjects that are close to the decision boundary. A post-hoc analysis is needed

to further investigate the predictive capacity of the covariates.

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